VANSH RAI SAINI

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Robotics engineer (M.S. ECE, UC San Diego) building **vision-language-action models**, **vision-language navigation**, and learning-based manipulation, plus 2 years shipping multi-robot planning and control software for warehouse AGV fleets.

EDUCATION

University of California, San Diego — M.S. ECE (Robotics) | CGPA: 3.8/4 **2025–2027**
Coursework: Probabilistic AI, Planning & Learning in Robotics, Sensing & Estimation in Robotics, Robot Manipulation

IIT Jammu — B.Tech. Mechanical Engineering (Minor: Computer Science) | CGPA: 8.47/10 **2019–2023**
Coursework: Computer Vision, Deep Learning, Machine Learning, Control Systems, Sensors & Actuators

RESEARCH EXPERIENCE

Existential Robotics Lab | UC San Diego (PI: Nikolay Atanasov) **Jan 2026 – Present**

- Built a **zero-shot vision-language navigation system** on a Jackal robot: a VLM converts natural-language instructions into **pixel waypoints** from history-conditioned multimodal prompts, including Intel RealSense images.
- Developing a **semantic-map-grounded planner** in which the VLM reasons over an accumulated semantic map, not just the current view, to select the next goal point and waypoint.
- **Deployed StreamVLN on a real Jackal robot**, fusing model-generated actions with live camera streams and temporal context; evaluated the model on 4× NVIDIA A10 GPUs.

Machine Learning Research Assistant | IIT Jammu (PI: Dr. Shaifu Gupta) **Jul 2022 – Mar 2023**

- Predicted critical-heat-flux location in annular pipes with SVM and deep learning (**R^2 = 0.94**); expanded the dataset with Wasserstein GANs and used PCA/mRMR for feature selection.
- **Published** in Progress in Nuclear Energy. [Paper]

PROJECTS

From-Scratch VLA Model for Robotic Manipulation | PyTorch, LeRobot, Flow Matching, LIBERO

- Built a **π0-style Vision-Language-Action model from scratch in PyTorch**: SmolVLM2 (256M) backbone with a custom **flow-matching transformer action head**; actions generated by iterative Euler integration at inference.
- Trained end-to-end through LeRobot on 432 LIBERO-Spatial demonstrations (96 GPU-hours), reaching an **80% success rate across all 10 LIBERO-Spatial tasks**.

Robotic Arm Control with Reinforcement Learning | PPO, Gazebo, ROS, TensorBoard

- Trained a 6-DoF manipulator with **PPO** in Gazebo for goal reaching; converged in 350 episodes (reward and policy loss tracked in TensorBoard); reached **17/20 randomized targets within 5 cm.**

Visual-Inertial SLAM | OpenCV, EKF

- Built a **stereo visual-inertial SLAM system with an Extended Kalman Filter** that fuses IMU data and stereo observations to jointly estimate robot pose and 3D landmarks.
- Implemented stereo feature matching and optical-flow tracking (OpenCV) for stable landmark correspondences, with an IMU-kinematics prediction step and a nonlinear stereo measurement model.

INDUSTRY EXPERIENCE

Unbox Robotics | Robotics Software Engineer | Pune, India **May 2023 – Jul 2025**

- Designed and deployed a **multi-agent path planner** (C++/Python) for AGV fleets with multi-cycle deadlock avoidance, keeping swarm traffic flowing continuously.
- Added a traffic-aware heuristic to A*, **increasing system throughput 6%** in high-density warehouse environments.
- Built a Gazebo + ROS 2 simulation with a pure-pursuit controller; wrote GoogleTest/pytest suites for faster debugging and safer releases.

SKILLS

Languages: Python, C++
Robot Learning: Reinforcement Learning (PPO), Imitation Learning, VLAs, Flow Matching, Diffusion Policies, World Models
Autonomy: Motion Planning, Multi-Agent Path Planning, SLAM, State Estimation, Optimization
Tools: PyTorch, LeRobot, ROS2, Isaac Sim, MuJoCo, Gazebo, OpenCV, Docker, Linux, GoogleTest, pytest